

Pair Coulomb Density Matrix on a Sphere

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(Dated: May 28, 2026)

We tabulate the diagonal pair Coulomb density matrix on a sphere.

Keywords: Pair Coulomb Density Matrix; Sphere; Table

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I. INTRODUCTION

We want to find the Pair Coulomb Density Matrix (PCDM) on a Sphere so to extend the work of Ref. [1–3].

We will denote with a Greek index a coordinate basis component and with a Roman index a particle label. We will adopt Einstein summation convention to tacitly assume a sum over repeated Greek indexes. We distinguish between upper contravariant Greek indexes and lower covariant Greek indexes. Passing between the two is accomplished with a contraction with the metric tensor, as usual.

The metric tensor of the Sphere of radius a is

$$||g_{\alpha\beta}|| = a^2 \begin{pmatrix} 1 & 0 \\ 0 & \cos^2 \theta \end{pmatrix}. \quad (1.1)$$

with $g(\mathbf{r}) = \det||g_{\alpha\beta}|| = a^4 \cos^2 \theta$.

It is important to realize that on a curved surface the two body problem is generally not separable into a center of mass and a relative one body problem. Therefore, in full generality, we will study the position representation $\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)$ of the density matrix $\rho = \exp(-Ht)$, where H is the Hamiltonian operator for the two bodies. These are initially at $\mathbf{r}_1$ and $\mathbf{r}_2$ and at an imaginary time t later they are at $\mathbf{r}'_1$ and $\mathbf{r}'_2$.

Therefore, the PCDM on the Sphere will satisfy the Bloch equation

$$\frac{\partial \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)}{\partial t} = \left(\frac{\lambda_1}{\sqrt{g_1}} \nabla_{1\alpha} \sqrt{g_1} \nabla_1^\alpha + \frac{\lambda_2}{\sqrt{g_2}} \nabla_{2\alpha} \sqrt{g_2} \nabla_2^\alpha - \phi(\mathbf{r}_1, \mathbf{r}_2) \right) \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t), \quad (1.2)$$

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; 0) = \delta(\mathbf{r}_1 - \mathbf{r}'_1) \delta(\mathbf{r}_2 - \mathbf{r}'_2) / \sqrt{g_1 g_2}, \quad (1.3)$$

where, for particle $i = 1, 2$, $\lambda_i = \hbar^2/2m_i$ with m_i the mass of particle i , $g_i = a^4 \sin^2 \theta_i$, $\nabla_i = (\nabla_{i1}, \nabla_{i2}) = (\partial/\partial\theta_i, \partial/\partial\varphi_i)$ is the gradient respect to the coordinates $\mathbf{r}_i = (r_i^1, r_i^2) = (\theta_i, \varphi_i)$, δ is a two dimensional Dirac delta function, ρ is the pair density matrix, and ϕ is the Coulomb pair potential.

The Coulomb pair potential for particles moving **on** the surface can be chosen as

$$\phi_{\text{on}}(\mathbf{r}_1, \mathbf{r}_2) = Z_1 Z_2 e^2 / d(\mathbf{r}_1, \mathbf{r}_2), \quad (1.4)$$

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where $Z_i e$ is the charge of particle i and for the distance between the two points $\mathbf{r}_1$ and $\mathbf{r}_2$, $d(\mathbf{r}_1, \mathbf{r}_2)$, on the Sphere we can either choose the geodesic distance

$$d_g(\mathbf{r}_1, \mathbf{r}_2) = a \arccos[\sin \theta_1 \sin \theta_2 + \cos \theta_1 \cos \theta_2 \cos(\varphi_1 - \varphi_2)], \quad (1.5)$$

or, viewing the Sphere as embedded in the three dimensional space, the Euclidean distance

$$d_e(\mathbf{r}_1, \mathbf{r}_2) = 2a \sin(d_g/2a). \quad (1.6)$$

For particles living **in** the surface [4] we may choose

$$\phi_{\text{in}}(\mathbf{r}_1, \mathbf{r}_2) = -Z_1 Z_2 e^2 \ln[d_e(\mathbf{r}_1, \mathbf{r}_2)/\ell], \quad (1.7)$$

with ℓ a constant with the dimensions of a distance. We will call ϕ_{on}^g , ϕ_{on}^e , ϕ_{in} the three cases just illustrated.

We will measure energy in units of the Hartree e^2/a_B , with e the charge of the electron and $a_B = \hbar^2/m_e e^2$ Bohr radius, with m_e the mass of the electron. And length in units of the Bohr radius a_B .

The free particles, $\phi \rightarrow 0$, pair density matrix can be determined from the work of Bastianelli and Corradini [5]. We will call their solution for the single particle i , $\rho_{i0}(\mathbf{r}_i, \mathbf{r}'_i; t)$. Therefore the pair free density matrix will be

$$\rho_0(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t) = \rho_{10}(\mathbf{r}_1, \mathbf{r}'_1; t) \rho_{20}(\mathbf{r}_2, \mathbf{r}'_2; t). \quad (1.8)$$

Then our PCDM can be written as

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t) = \rho_0(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t) e^{-P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)}, \quad (1.9)$$

where P is the *inter-action*.

Aim of this work is to tabulate $P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)$ for the three Coulomb cases illustrated above, P_{on}^g , P_{on}^e , P_{in} . For small $t \ll 1$, for each of these cases, the *primitive approximation* [6]

$$P_{\text{primitive}} = t\phi, \quad (1.10)$$

holds.

Therefore, introducing a timestep τ , we can find the PDCM at all subsequent timesteps through the following convolution

$$\begin{aligned} \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; n\tau) = & \int \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_{1,1}, \mathbf{r}_{2,1}; \tau) \rho(\mathbf{r}_{1,1}, \mathbf{r}_{2,1} | \mathbf{r}_{1,2}, \mathbf{r}_{2,2}; \tau) \cdots \rho(\mathbf{r}_{1,n-1}, \mathbf{r}_{2,n-1} | \mathbf{r}'_1, \mathbf{r}'_2; \tau) \times \\ & \prod_{k=1}^{n-1} \sqrt{g_{1,k} g_{2,k}} d\mathbf{r}_{1,k} d\mathbf{r}_{2,k}, \end{aligned} \quad (1.11)$$

where we denote with $\mathbf{r}_{i,k}$ the position of particle i at timeslice k , $g_{i,k}$ is the determinant g calculated at $\mathbf{r}_{i,k}$, $d\mathbf{r} = d\theta d\varphi$, and we choose τ small enough so that

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; \tau) \approx \rho_0(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; \tau) e^{-\tau[\phi(\mathbf{r}_1, \mathbf{r}_2) + \phi(\mathbf{r}'_1, \mathbf{r}'_2)]/2} \quad (1.12)$$

$$= \rho_{10}(\mathbf{r}_1, \mathbf{r}'_1; \tau) \rho_{20}(\mathbf{r}_2, \mathbf{r}'_2; \tau) e^{-\tau[\phi(\mathbf{r}_1, \mathbf{r}_2) + \phi(\mathbf{r}'_1, \mathbf{r}'_2)]/2}, \quad (1.13)$$

is a good approximation. According to the analysis of Ref. [5] we can further approximate

$$g^{1/4}(\mathbf{r}_i) \rho_{i0}(\mathbf{r}_i, \mathbf{r}'_i; \tau) g^{1/4}(\mathbf{r}'_i) \approx \exp \left\{ -\frac{1}{2\mu_i \tau} \delta_{\alpha\beta} (\mathbf{r}_i - \mathbf{r}'_i)^\alpha (\mathbf{r}_i - \mathbf{r}'_i)^\beta - \tau \frac{V_R(x_i) + V_R(x'_i)}{2} \right\}, \quad (1.14)$$

where $\delta_{\alpha\beta}$ is the Kronecker delta, $x_i = \delta_{\alpha\beta} \mathbf{r}_i^\alpha \mathbf{r}_i^\beta$, and V_R is the effective potential due to the curvature of the Sphere, $R = 2M^2$ with $M = 1/a$, namely

$$V_R(x) = -\frac{M^2}{6} - \frac{5(Mx)^2 - 3 + [(Mx)^2 + 3] \cos(2Mx)}{48x^2 \sin^2(Mx)}, \quad (1.15)$$

which is equation (2.32) of Ref. [5]. We will perform the $4(n-1)$ integrations in Eq. (1.11) through a simple quadrature method.

Even if in general we will be interested in the full $P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)$ in practice in a many body simulation we may use the *end point approximation* for the inter-action as shown in section IV.F of Ref. [6] where one only needs the diagonal components $P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; t) = P(d_g(\mathbf{r}_1, \mathbf{r}_2); t)$. Where on a sphere the geodesic distance $d_g(\mathbf{r}_1, \mathbf{r}_2)$ between any two points $\mathbf{r}_1$ and $\mathbf{r}_2$ is bounded from above by πa . Due to the Sphere symmetry, without loss of generality, we may choose $\mathbf{r}_1 = \mathbf{0}$ and $\mathbf{r}_2 = (\theta, 0)$, so that $d_g = a\theta$.

On the diagonal we will have

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; t) = \rho(d_g(\mathbf{r}_1, \mathbf{r}_2); t) = \rho_{10}(\mathbf{0}, \mathbf{0}; t) \rho_{20}(\mathbf{0}, \mathbf{0}; t) e^{-P(d_g(\mathbf{r}_1, \mathbf{r}_2); t)}, \quad (1.16)$$

and $\ln[2\pi t \sqrt{g(\mathbf{0})} \rho_{i0}(\mathbf{0}, \mathbf{0}; t)]$ is given up to orders t^8 in Ref. [5] (see their Eq. (3.23)). For completeness we report their perturbative expansion in Eq. (A1) of Appendix A.

We will then tabulate

$$P(m\delta; n\tau) = -\ln \left[\frac{\rho(m\delta; n\tau)}{\rho_{10}(\mathbf{0}, \mathbf{0}; n\tau) \rho_{20}(\mathbf{0}, \mathbf{0}; n\tau)} \right], \quad n = 1, 2, 3, \dots, \quad m = 1, 2, 3, \dots, m_{\max} \quad (1.17)$$

for given small τ and δ so that $m_{\max} = \pi a / \delta$. For an imaginary time $t_n = n\tau$ this will require a number of $4(n-1)m_{\max}$ quadratures to calculate $\rho(m\delta; n\tau) = \rho(d_g(\mathbf{r}_1, \mathbf{r}_2); t_n) = \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; t_n)$ for $m = 1, 2, 3, \dots, m_{\max}$, through the convolution (1.11). Each convolution requires 2 quadratures over the azimuthal angle $\varphi \in [-\pi, \pi]$ and the polar angle $\theta \in [-\pi/2, \pi/2]$. Choosing an angular discretization $\delta_\varphi = 2\delta_\theta$ we will have that each angular integral reduces to a sum over $l = \pi/\delta_\theta$ terms. Then in order to tabulate $P(d; t_n)$ we need $l^{4(n-1)}$ for each m functions evaluations. This becomes more computationally demanding at later imaginary times t but in a many body simulation the pair-product inter-action is needed at a fixed small t , which determines the discretization error [6]. So one needs to tabulate it just once, at the beginning of the simulation.

In Table I we show $P(m\delta; n\tau)$ for $a = 1, m_1 = m_2 = m_e, Z_1 = -Z_2 = -1, \tau = 0.001, \delta = 0.1$ and $n = 10$.

TABLE I. Table for $P(m\delta; n\tau)$ for $a = 1, m_1 = m_2 = m_e, Z_1 = -Z_2 = -1, \tau = 0.001, \delta = 0.1$ and $n = 10$. For the quadratures we used Simpson rule with $\delta_\theta = 0.01$ so that $l = 314$.

P_{on}^g	P_{on}^e	P_{in}	m
			1
			2
			20
			31

II. THE CODE

We give here the FORTRAN code listing to generate the table of Eq. (1.17):

Appendix A: $\rho_{i0}(\mathbf{0}, \mathbf{0}; t)$

Calling $\mu_i = m_i/m_e$ and $R = 2/a^2$ the Sphere scalar curvature, we find from equation (3.23) of Ref. [5]

$$\ln \left[2\pi t \sqrt{g(\mathbf{0})} \rho_{i,0}(\mathbf{0}, \mathbf{0}; t) \right] = \frac{tR_i}{12} + \frac{(tR_i)^2}{6!} \frac{1}{2} + \frac{(tR_i)^3}{9!} 16 + \frac{(tR_i)^4}{10!} \frac{59}{4} + \frac{(tR_i)^5}{11!} \frac{58}{3} + \frac{(tR_i)^6}{13!} \frac{550789}{1260} + \frac{(tR_i)^7}{14!} \frac{46838}{45} + \frac{(tR_i)^8}{17!} \frac{596004707}{720} + \mathcal{O}(t^9), \quad (A1)$$

where $R_i = R/\mu_i$.

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Funding

None declared.

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